

Do Democrats or Republicans Experience More Difficulty Voting?

Data 203: Lab 1 Part 2

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1 Introduction

Given the extremely polarized political climate that we experience today in 2023, there are so many factors that go into how national elections take place in the United States. In the past decade, national elections have been decided amidst very close margins, making it an increasingly competitive process that depends heavily on voter turnout. Voter turnout is heavily influenced by voter motivation, as well as feelings like anger and fear concerning diverse political issues and more importantly, difficulties faced during the voting process. The disparities in voter turnout among political parties, stemming from these difficulties, play a pivotal role in determining election outcomes. This context drives our analysis, leading us to investigate:

Do Democratic voters or Republican voters experience more difficulty voting?

The answer of this pivotal inquiry holds the promise of illuminating pathways for upcoming political campaigns, delineating strategies for specific parties to amplify the electoral engagement of their followers. Beyond mere tactical advantages, it could provide constructive critiques that might prompt governmental refinements, ensuring a more fluid and equitable voting milieu. More profoundly, a discerning comprehension of the voting challenges peculiar to each faction could shed light on the roots of our nation's political polarization, potentially offering a tapestry of understanding that could guide our endeavors to bridge this profound divide.

As part of our report, we will be conceptualizing the research question in terms of data so we can visualize who a voter is, who a Democrat or Republican is, as well as what constitutes difficulty voting. Next, we plan to clean the data to effectively capture the most effective variables which we will then use to create plots and visualizations regarding voter information. Based on our findings, we will perform hypothesis testing to yield significant results.

2 Data and Methodology

Our data is sourced from the 2018 American National Election Studies (ANES). Conceptualizing our research question using the data at hand requires clarifying specific concepts we wish to use. When evaluating who or what a voter is, we define an individual to be a voter if they have voted in either of the past two elections. We have done so by filtering the ANES data using the **turnout20** variable, which asked respondents if they voted in the 2020 election, and the **turnout22** variable which asked respondents if they voted in the 2022 election. Our filtering process was done by retaining rows when **turnout20 == 1**, stating that an individual voted in the 2020 election, and **turnout22 == 1 or 2 or 3**, which represents individuals who definitely voted in person on November 8th, before November 8th, or by mail in the 2022 election respectively.

We then had to define whether a voter was a Republican or Democrat. To do so, we used the **pid_x** variable which was a summary of an individual's party ID on a scale from 1-7, 1 being strong democrat and 7 being strong republican. For simplicity of our analysis, we classified anyone who was somewhat leaning towards Democrat, which was when **pid_x = 1, 2, or 3**, representing strong Democrat, Democrat, and lean Democrat, respectively, a Democrat, and classified anyone who was somewhat leaning towards Republican, which was when, **pid_x = 5, 6, or 7** representing, lean Republican, Republican, and strong Republican, respectively, a Republican. In addition, we left out individuals classified themselves as Independent, **pid_x = 4**, out of our analysis.

Finally, we had to define our measure of difficulty voting. To do so we took the variables **votehard**, which was a scale from -1 to 5 answering the question, "How difficult was it for you to vote?" (-1 being that they skipped the question, 1 being that it was not difficult at all, and 5 being that it was extremely difficult), as well as all of the binary variables **vharder_0, vharder_1, ..., vharder_11**, which address specific reasons that an individual may have difficulty voting (e.g. difficulty registering, long wait time, hard to figure out my polling place, etc.). We also pulled in **vharder_12**, which was a binary variable that asked if the individual had no difficulty at any capacity. We created a new variable named **difficulty**, which concatenates the information provided by the **votehard** and **vharder** variables. We wanted to account for the people who skipped the **votehard** question because they were still valuable individuals in our research because they were indeed voters with a specific party. Therefore, for the individuals who had a **votehard = -1**, we first decided to inspect their answers for the **vharder_12** variable. If an individual had a **votehard = -1**, and did select

vharder_12 (**vharder_12** = 1), then we assign a value of 1 to the **difficulty** variable, indicating that the individual did not have any difficulty with the voting process. In any other case where the individual had a **votehard** = -1, we created a function that would loop through **vharder_0** to **vharder_11** columns of our data and calculate the sum of the responses of those columns. Although the numerical representations we created of the voter difficulty based on the **vharder_0** to **vharder_11** variables is based on the number of total difficulties selected by the individual, the final **difficulty** variable was subjectively created based on our interpretation of these variables. If either one or two of the **vharder_(0-11)** questions were selected, we assigned the **difficulty** to be equal to 2, meaning it was a little difficult for the individual to vote. If the voter selected three or four of the **vharder_(0-11)** questions, we assigned the **difficulty** to be 3, meaning voting was moderately difficult. If five or six of the **vharder_(0-11)** questions were selected, we assigned the **difficulty** to be 4, or very difficult. And finally, if more than 6 **vharder_(0-11)** questions were selected, we assigned the **difficulty** to be equal to 5, indicating that it was extremely difficult for the individual to vote. This **difficulty** variable is what we ultimately used to conduct our analysis.

By cleaning the ANES dataset into a more refined subset, we created a conclusive data frame encompassing both Republican and Democratic voters, each of whom interacted with specific measures of voting difficulty. This encompassed voters from both the 2020 and 2022 elections, detailing not only the general degree of voting challenges they faced but also specific issues they encountered. With this refined data in hand, we further delineated it into two distinct tables: one dedicated to Democratic voters and the other exclusively for Republican voters.

In order to visually compare the proportion of voter difficulties experienced by both the Democrat and Republican parties, we created a side by side bar chart depicted in **Figure 1**.

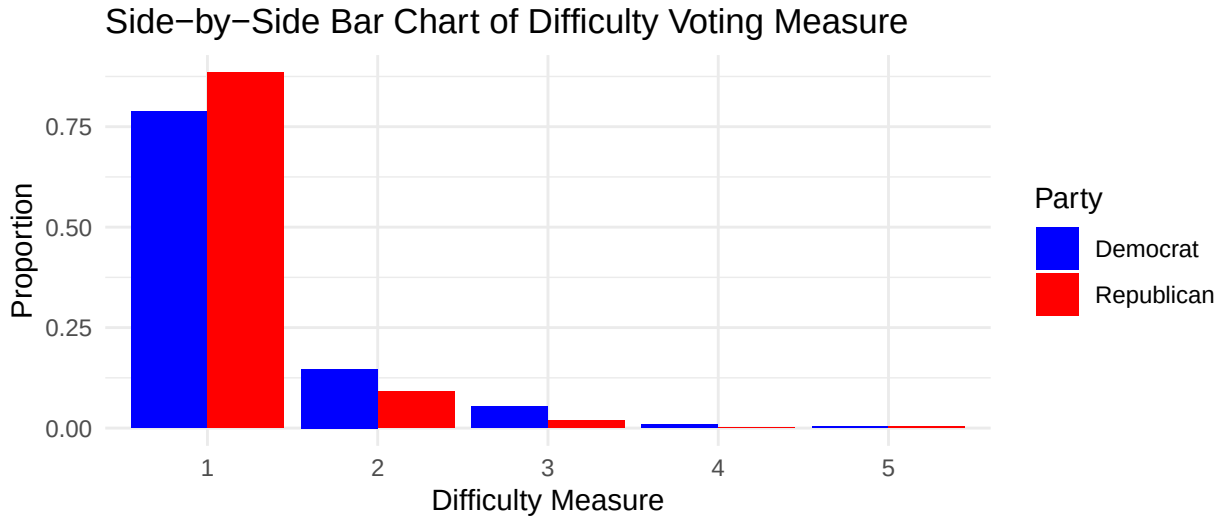


Figure 1: Proportion of Difficultly Voting for Both Democrat and Republican Voters.

We can see, at a difficulty measure of 1 (Not difficult at all), the proportion of Republican voters that experienced no difficulties at all was much higher than that of Democrat voters. In addition, as the difficulty increases, there is a higher proportion of Democratic voters that experience these difficulty measures than that of Republican voters, suggesting that Democrats are experiencing a more difficult time voting than Republicans. A plausible interpretation of the figure suggests that while a predominant segment of voters from both affiliations encountered negligible voting barriers, Republican voters, on average, might experience a relatively less difficult voting process.

3 Hypothesis Testing

To compare the voting difficulty between Democrats and Republicans, our null hypothesis can be phrased as follows:

Null Hypothesis: *The difficulty of voting for Democrats is equal to the difficulty of voting for Republicans*

In order for a Wilcoxon rank sum test to produce reliable inference, the variable of interest, difficulty of voting, must be ordinal, and the sample must be independent and identically distributed. Voter difficulty is an ordinal variable by nature, as values assigned to voter difficulty do range from less difficult to more difficult. However the differences between categories in ordinal variables are not necessarily consistent between intervals, so the variable is not metric.

According to the ANES pilot 2022 questionnaire, individuals are sampled from a diverse set of respondents who voluntarily completed the survey. There is a possibility that the voluntary nature of the survey introduces dependencies, however sample matching based on gender, age, race, and education is used to make sure individuals who complete the survey represent the population, which may decrease possibilities of bias. After controlling for the matching and weighting factors, the participation or non-participation in the survey is not systematically related to any of the other survey questions, according to the ANES 2022 User's Guide and Codebook. Therefore, the sample distribution of survey variables will approximately match the distribution of those variables in the population, so the data may be used for population inference. We can say that individual voters are independent of one another, as they vote in private and have limited contact with other individuals during the voting process. Any dependencies that we might observe due to family ties, geographic location, etc., would not impact our assumption of independence among voters given the size of the sample.

Additionally, our test is unpaired, as our groups of interest (Democrats and Republicans) are not related or matched in any way within the data. Therefore, this further meets our requirements for using the Wilcoxon rank sum test, which is a non-parametric, unpaired test, to compare Democrats and Republicans to determine if there is a statistically significant difference between them.

4 Test, Results and Interpretation

```
test <- wilcox.test(Democrat$difficulty, Republican$difficulty, alternative = "two.sided")
```

The results of the Wilcoxon rank sum test suggest that there is enough evidence to reject the null hypothesis, as the p-value of 1.358e-05 was less than our significance level of 5%. Therefore, the test yields evidence that there is a significant difference in voter difficulty between Democrats and Republicans.

In order to check for the effect size and practical significance of this difference, we must calculate a biserial Spearman rank correlation for the effect size:

```
practical_significance <- cor.test(voters$difficulty, voters$Party, method = "spearman")
```

The estimated rho value is approximately -0.131, which indicates a weak negative correlation between party and voter difficulty. As we assigned Democrats a value of 0 and Republicans a value of 1 in our dataset, this indicates that Republicans tend to have lower values of voter difficulty compared to Democrats.

The small observed p-value suggests that the relationship is unlikely to have occurred by random chance. The practical significance of this fairly weak negative difference between the variables would require further investigation in regards to the presence of other variables. These results are important in addressing our goal of maximizing voter turnout midst close margins in elections.

Findings from this study may aid campaigners and policymakers in directing resources to support individuals who experience more difficulty in voting. Campaigners and policymakers can use the study's findings to design targeted outreach efforts aimed at addressing barriers that underrepresented voter groups may experience. Voter education can also be utilized to help individuals understand the voting process. Additionally, policy reforms, related to voting, may create a more convenient voting environment, such as extending voting hours, improving accessibility for non-English speakers, etc. can help bridge gaps and decrease polarization. These initiatives would likely facilitate broader civic engagement and voting practices.